Distance, Mobility, and Mode Choice: A Spatial Analysis of School Travel Behaviour in Ireland

Ollscoil na Gaillimhe
University of Galway

Judith Beltrán López, Sananthaa Jagadesan Senthamaraikannan[1]

Background and Problem

- Daily travel to school significantly influences physical activity levels in children and adolescents
- Increased use of unhealthy transport has reduced walking and cycling
- Active travel provides important opportunities for regular physical activity
- In Ireland, school location, distance to school, and neighbourhood features affect travel choices
- Travel habits formed in childhood often continue into adulthood

Objectives of Project

1. Map and analyse the spatial distribution of schools across Ireland.
2. Examine the distance between students' residential locations and their schools, and investigate student mobility related to school attendance.
3. Analyse school travel behaviour and assess the prevalence of healthy versus less healthy commuting modes.
4. Explore the relationship between school proximity, travel mode, and potential health implications.

Data Sources and Datasets

Data on schools in Ireland are of the following Types:

- School-Level Data (Department of Education): School type, enrolment, and geographic location (Eircode,latitude, longitude).
- Settlement-Level Travel Data (CSO): Census data on modes of travel to education at settlement level across multiple years.
- Spatial Boundary Data (CSO): Geographic boundaries for towns and urban areas used for spatial mapping and analysis.

Early Results

The analysis began with an exploratory examination of the spatial distribution of schools across Ireland using maps and density plots. This helped validate the dataset, reveal spatial patterns, and highlight regional differences in school availability. The results show a clear urban–rural divide, with Dublin emerging as the main concentration of schools, suggesting shorter travel distances and greater potential for active transport, while rural areas are more likely to depend on motorised transport due to longer distances.

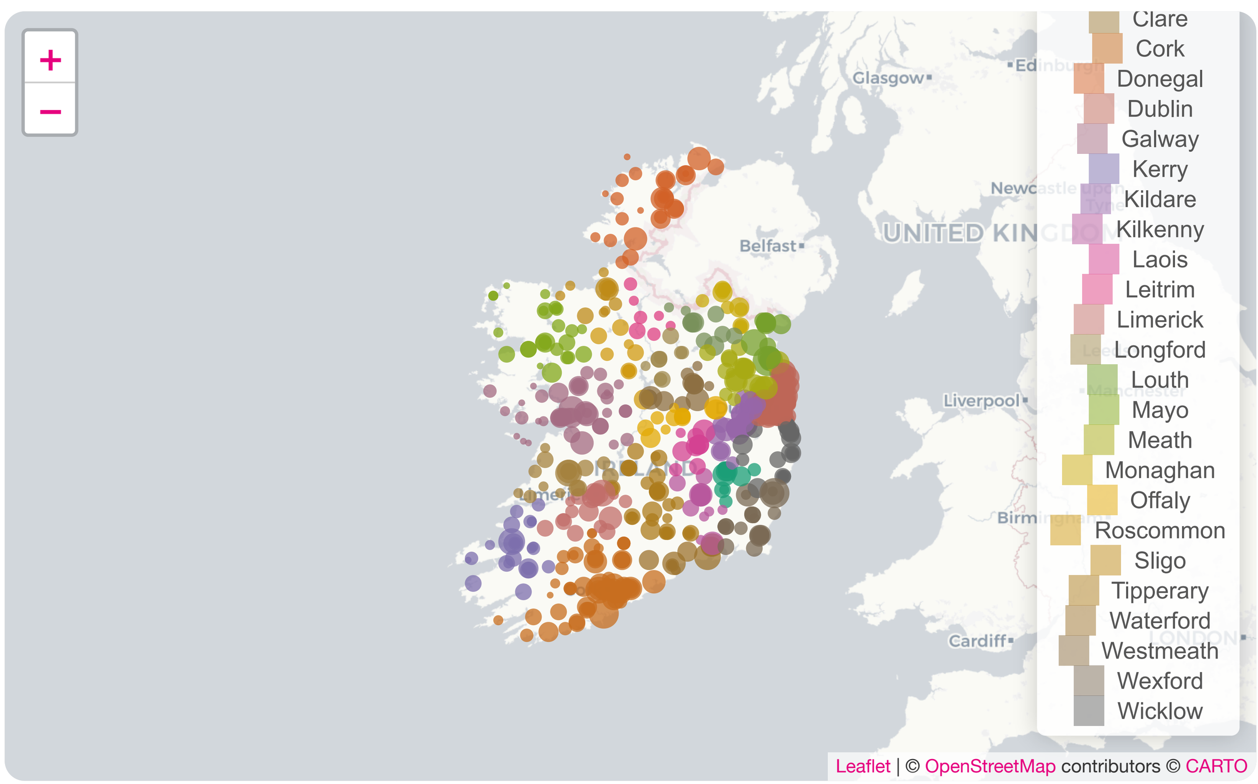

Figure 1: Density plots of Schools in Ireland

Kriging is a geostatistical interpolation technique that uses spatial relationships to estimate values at unsampled locations, helping to reveal spatial patterns and inequalities in school locations, travel behaviour, and related health impacts.

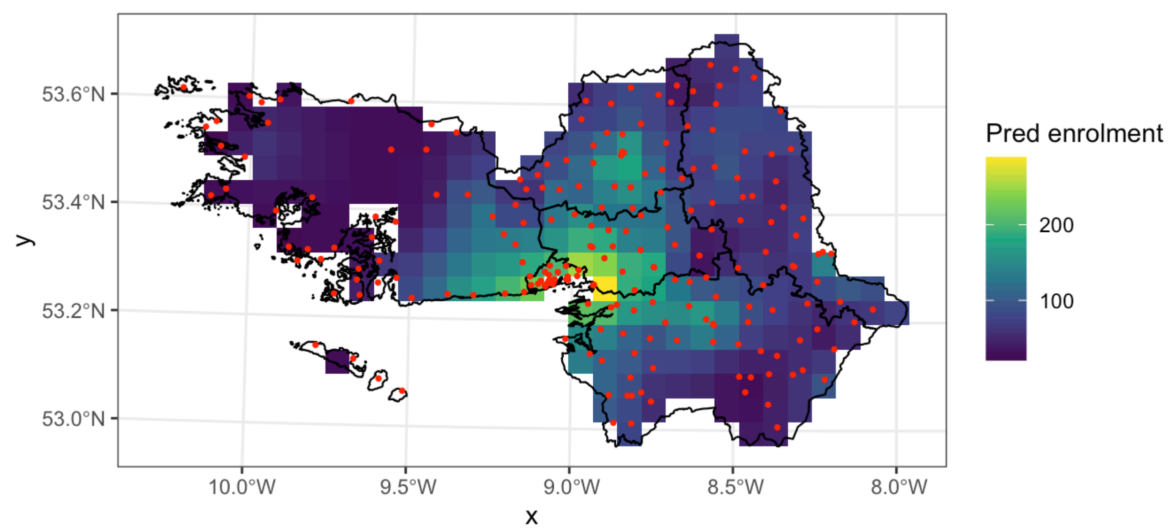

Figure 2: Kriging schools in Co Galway

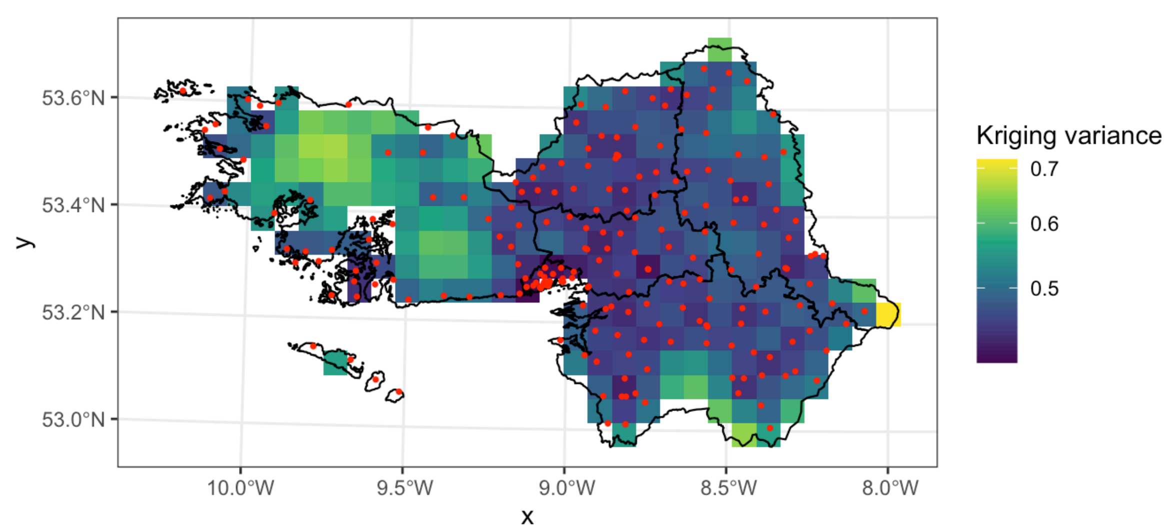

Figure 3: Kriging schools in Co Galway

This plot shows how total active transport use (walking and cycling) is distributed across towns. A logarithmic scale is used to account for large differences in town size.

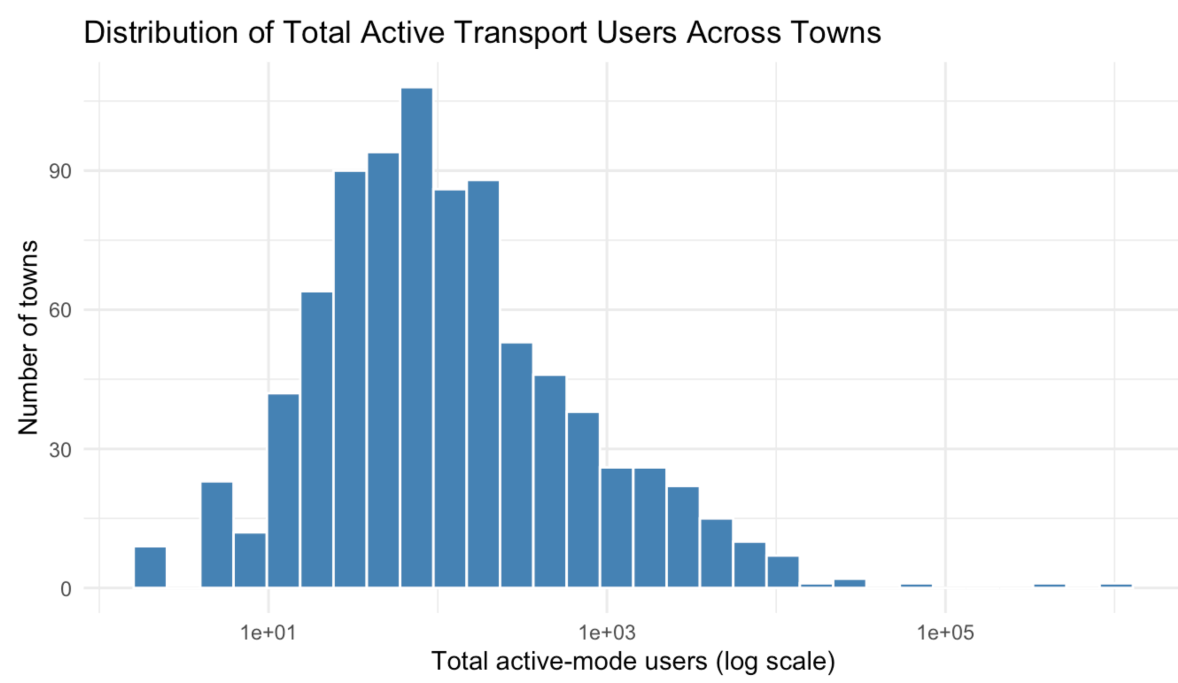

Figure 4: Histogram of active transport in Ireland

Next Project Steps

- Define outcome variables (e.g. enrolment size, accessibility, resource availability)
- Control for confounding factors (socioeconomic context, location, school size, demographics)
- Apply county-level kriging to examine spatial variation in school enrolment
- Extend analysis using regression or multivariable models to assess relationships between distance, travel mode, and school characteristics

GitHub

This project can be viewed at our GitHub repository here: https://github.com/judithbeltranlopez/Final-Project

References